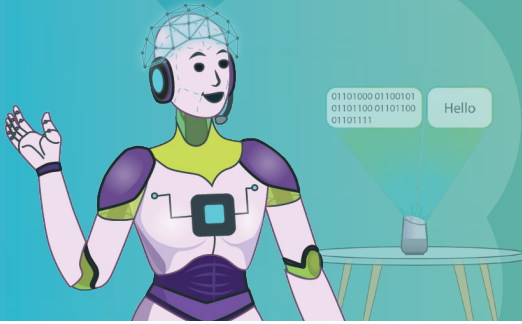


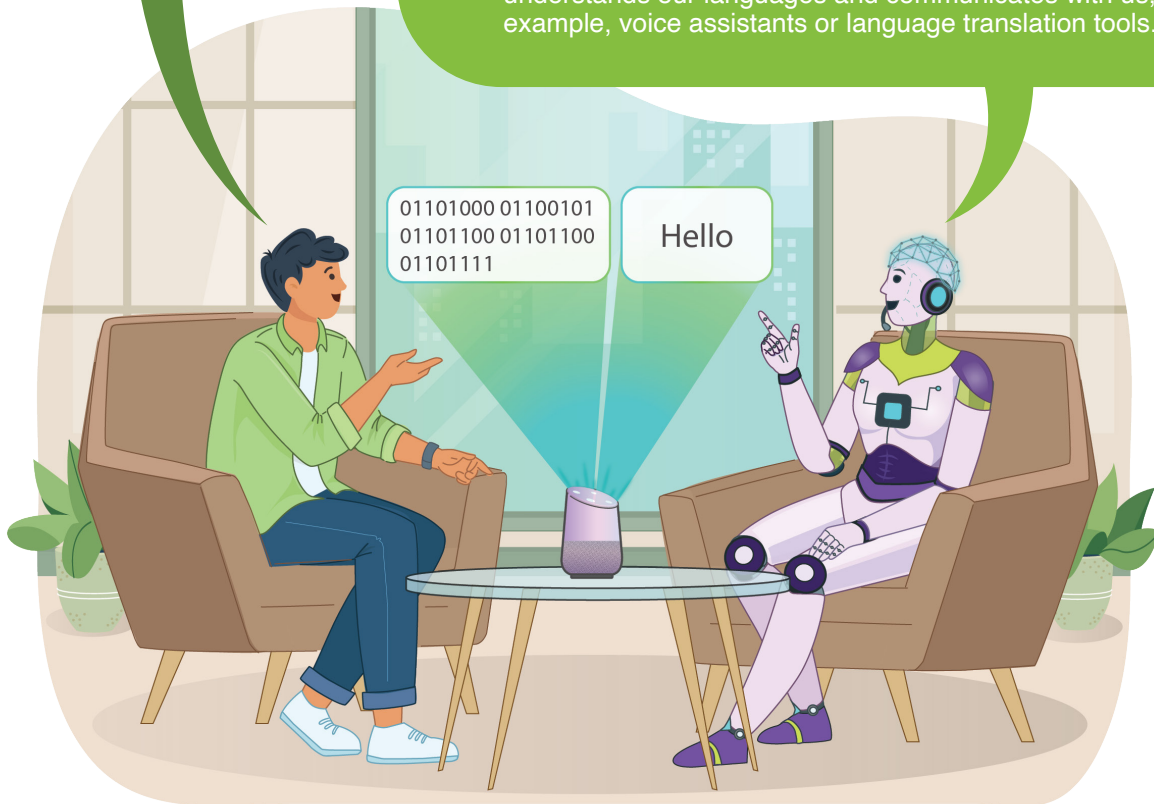
Introduction to NLP

08



Hi Mirai! How do computers understand what we're saying and reply to us in our language?

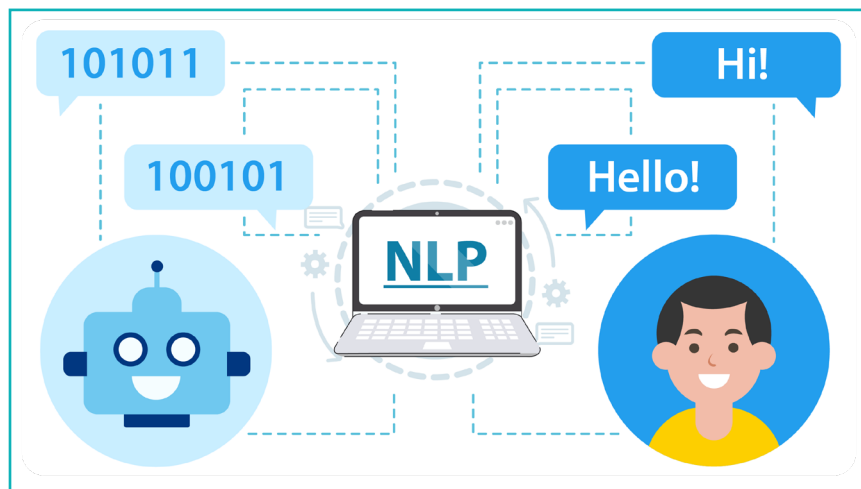
Computers have a special ability called Natural Language Processing (NLP). It is like having a super-smart friend who understands our languages and communicates with us, for example, voice assistants or language translation tools.



Language is extremely important in our daily lives. Languages are used for communication because they help us understand each other. They help us learn new things, express ourselves, form connections, and create a more connected and inclusive society.

Similarly, we can communicate with machines as well using a language. Machines can perform tasks that we ask them to do. However, communicating with machines itself was not an easy task. Machines like computers and mobile phones understand only machine language, i.e., binary language of 0s and 1s. It is very difficult for us humans to understand and communicate using binary language. To overcome this problem and make human-machine communication easy and effective, Natural Language Processing (NLP) is used.

NLP enables computers to read text, hear speech, understand it, and give an appropriate reply in the same language.



Imagine you are planning a vacation in a foreign country but you do not speak the local language. With a device that understands human language, you can use a translation app on your phone to communicate with the locals. You simply speak or type your question or request, and the app instantly translates it into the local language. This allows you to ask for directions, order food, or interact with the locals effortlessly.

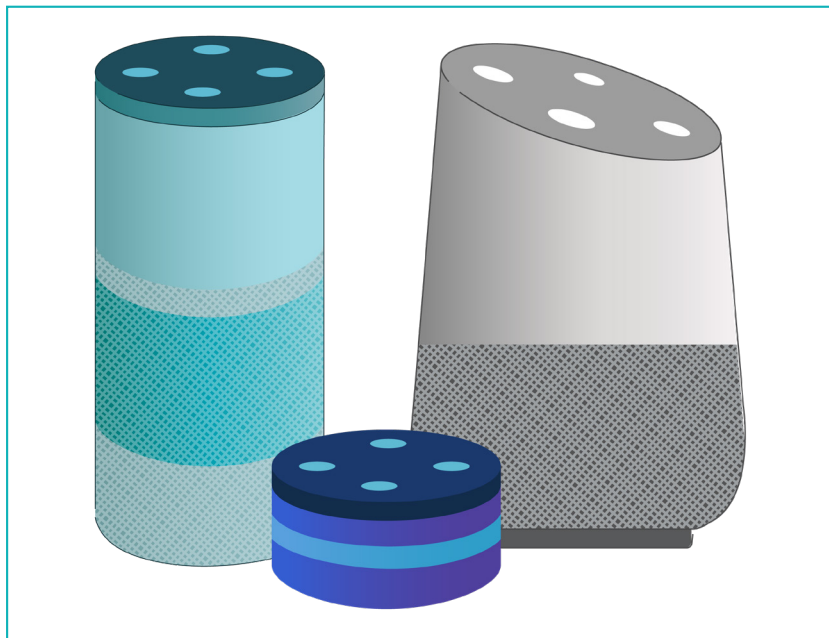


The machine's language understanding capability makes the communication process easier and more efficient. It helps in navigating the foreign environment with confidence and enjoying a more immersive experience.

In today's times, technology is all around us and it helps us in so many ways. We use computers, phones, and other cool gadgets to do things that used to be harder. One amazing thing about technology is that it can understand how we talk and write. It can listen to our words and understand what we mean. This makes it easier for us to communicate with our devices and get things done.

Words can have different meanings depending on the situation or the person. This is where Natural Language Processing (NLP) comes in. NLP is a field of Artificial Intelligence (AI) that focusses on the interaction between computers and human languages. It teaches computers how to understand and use language like humans do. It helps them to search more effectively, communicate with us in a smart way, and find important information in large amounts of text. It is like teaching computers to be language experts!

We typically get a virtual assistant like Google Assistant or Siri on a mobile phone. You can talk to them and ask them to find things for you on the Internet. For example, you can say "Hey Google, find videos about the solar system on YouTube." The assistant will find videos about the solar system and tell you their titles, how long they are, and who uploaded them. You can choose a video based on this information.



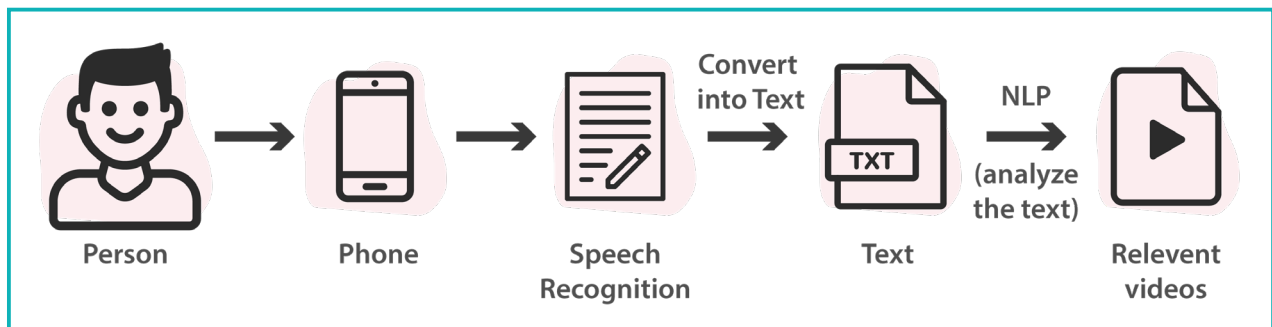
When you find a video you like, you can ask the virtual assistant to play it by saying something like "Play the video called 'Exploring the Solar System'." The assistant will communicate with YouTube and play the video on your device. You can watch the video and learn about the planets, their features, and their location in the solar system. The video might also explain things like gravity, orbits, and space exploration.

If you have more questions or want to learn more about a specific topic, you can ask the virtual assistant for more information or suggest related videos. This helps to expand your knowledge and find useful information for things like school assignments, understanding complex ideas, and getting ready for exams.

Have you ever thought about how it works?

When you talk to a device or app using your voice, the first thing it does is listen to what you say and convert it into printed words. This is called 'Speech Recognition Technology'. After that, the words are sent to the servers of the service you are using, like YouTube.

At the servers, special algorithms of Natural Language Processing (NLP) come into play. These algorithms are designed to understand human language. They carefully look at the written words and try to figure out what you mean and what you want.



NLP algorithms also try to understand the feelings and situation behind your words. By studying the words and phrases you used, the algorithms can tell if you are feeling happy, sad, or something else. This allows them to provide you with search results that are more relevant to your needs.

Once the system understands what you are looking for, it leverages that knowledge to discover the best videos, channels, or playlists from YouTube. YouTube offers a huge collection of videos, it helps in selecting the ones that are most suitable and related to what you want.

Definition of NLP

'Natural Language Processing' (NLP) is a field of AI that deals with the ability of computers to understand, interpret, and generate human language. It involves teaching computers to read, listen, speak, and write in ways that are similar to how people do. It enables computers to understand what we say or write and generates meaningful and useful responses.

In short, Natural Language Processing is concerned with the interaction of computers and humans using natural language.



In this voice search on YouTube, Artificial Intelligence (AI), and Machine Learning (ML) also have important roles to play. Let us take a closer look at what they do.

AI (Artificial Intelligence) - Artificial Intelligence (AI) is the method of teaching a computer machine to think and learn as humans do. AI is capable of learning from examples, make decisions, and even understand and respond to human language.

Example: In this YouTube voice search, the technology that understands and writes down what you say is called Speech Recognition. It is a kind of Artificial Intelligence (AI) that can recognize and process spoken words.

ML (Machine Learning) - Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables computers to learn from data and make predictions or decisions without explicit programming.

Example: In this voice search on YouTube, special computer programs called ML algorithms create language models by looking at the information you provide. These models learn how words and sentences are related to each other to better understand and make sense of your questions in a natural way.

Exercise

 Tick mark ☒ the correct answer

- 1 What is the role of ML algorithms in an AI chatbot?
 - a Training the chatbot to understand and respond to user queries ☐
 - b. Converting voice input into text ☐
 - c. Analysing sentiment in text ☐
 - d. Generating human-like text responses ☐
- 2 Which technology focusses on teaching computers to think and learn as humans do?
 - a. AI (Artificial Intelligence) ☐
 - b. ML (Machine Learning) ☐
 - c. NLP (Natural Language Processing) ☐
 - d. Speech Recognition ☐

Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) are three important fields that help computers understand and use human language.

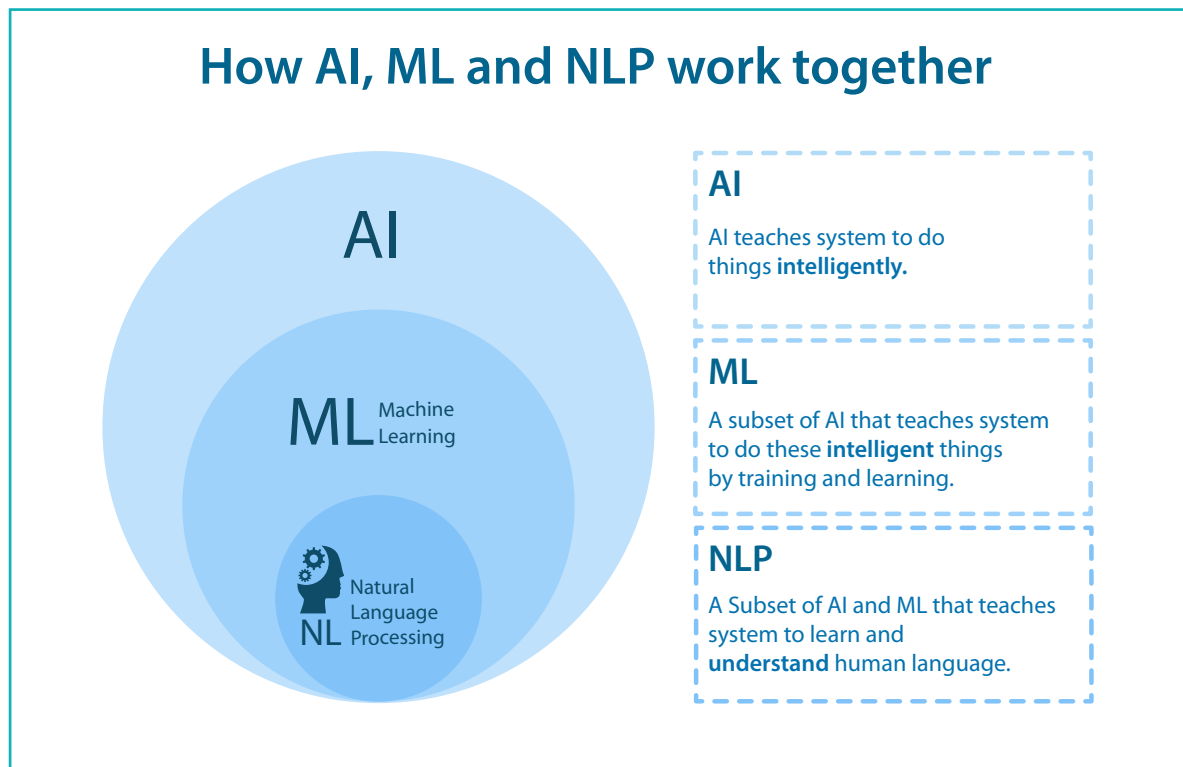
Let us see how they work together when you use voice search on YouTube.

In YouTube voice search, Machine Learning (ML) algorithms examine a large amount of text data from videos, comments, and other information on YouTube. They learn patterns and rules of how language is used on the platform.

The knowledge gained from studying this text data is used to create Natural Language Processing (NLP) models. These models are like special programs that understand human language. They can recognize words, understand sentences, and find meaning in the text.

Now, an AI system comes in. It uses the NLP models created with ML techniques. It takes your voice and converts it into text using Speech Recognition Technology. Then, it uses the NLP models to understand what you mean, how you feel, and what you want to find.

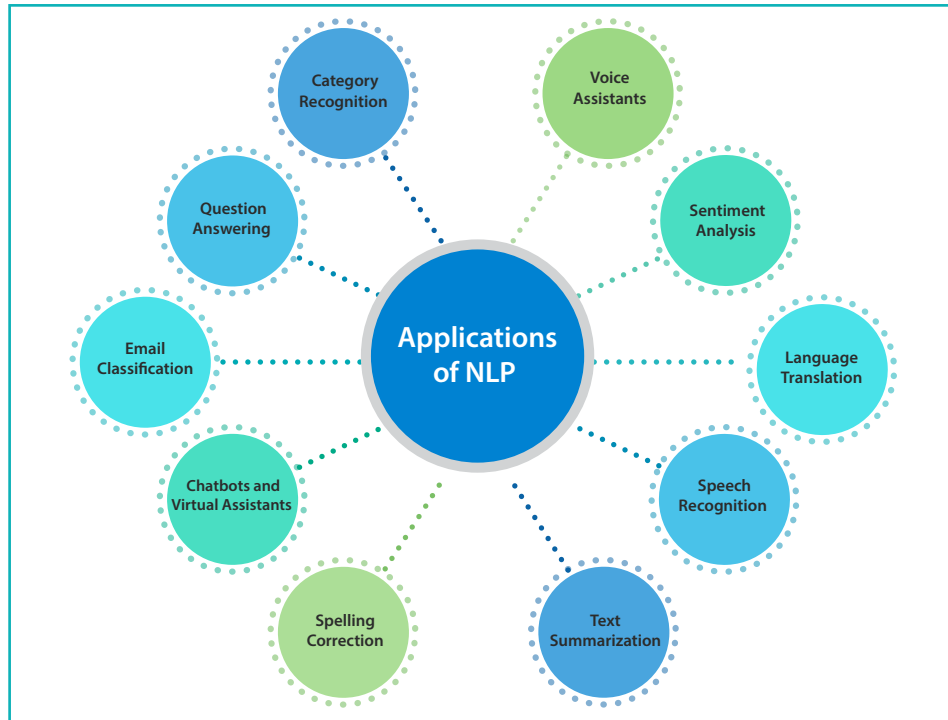
With this understanding, the AI system can find the most relevant videos, channels, or playlists on YouTube. It knows which ones match your search and gives you accurate results.



The ability of Natural Language Processing (NLP) to understand and create human language has a big impact on various industries such as healthcare, finance, and customer service.

Application of NLP

Natural Language Processing (NLP) is widely utilized in various fields, such as search engine optimization, text classification, chatbots and virtual assistants, machine translation, named entity recognition, healthcare research, fraud detection, cybersecurity, and content generation. Some most used applications are as below:



Some popular apps that use Natural Language Processing (NLP) techniques are as below:

- Voice assistants like Alexa, Siri
- Translation systems like Google Translate, IBM Watson
- Text analysis applications like Grammarly, Google Docs
- Search engines like Google, Bing

Bytes

- **Speech recognition technology** takes audio input of spoken words and converts it to text.
- **Virtual Personal Assistant (VPA)** is a software-based assistant that understands and responds to human queries using AI and NLP.
- **NLP models** are like intelligent computer programs that understand and interpret human language. It is designed to recognize words, understand sentence structures, and extract meaning from text.

Exercise

Fill in the Blanks.

- 3 AI, ML, and NLP work together to help computers understand and generate _____ language.
- 4 NLP helps in understanding the _____ and context of the query.

Important terminologies

In the world of NLP, there are important words and concepts that are essential for improving how we understand and work with human language. Some of them are explained as follows:

Sentence Segmentation: 'Sentence segmentation' is the process of dividing a text into individual sentences. It helps identify where one sentence ends and another begins, so that each sentence can be analyzed separately.

Word Tokenization: 'Word tokenization' involves breaking a text into individual words or tokens. It helps define the boundaries between words so that each word can be analyzed independently.

Part-Of-Speech (POS) Tagging: This is the process of labelling each word in a text with its corresponding part of speech, such as nouns, verbs, adjectives, and so on. It helps understand the role and function of each word in the sentence.

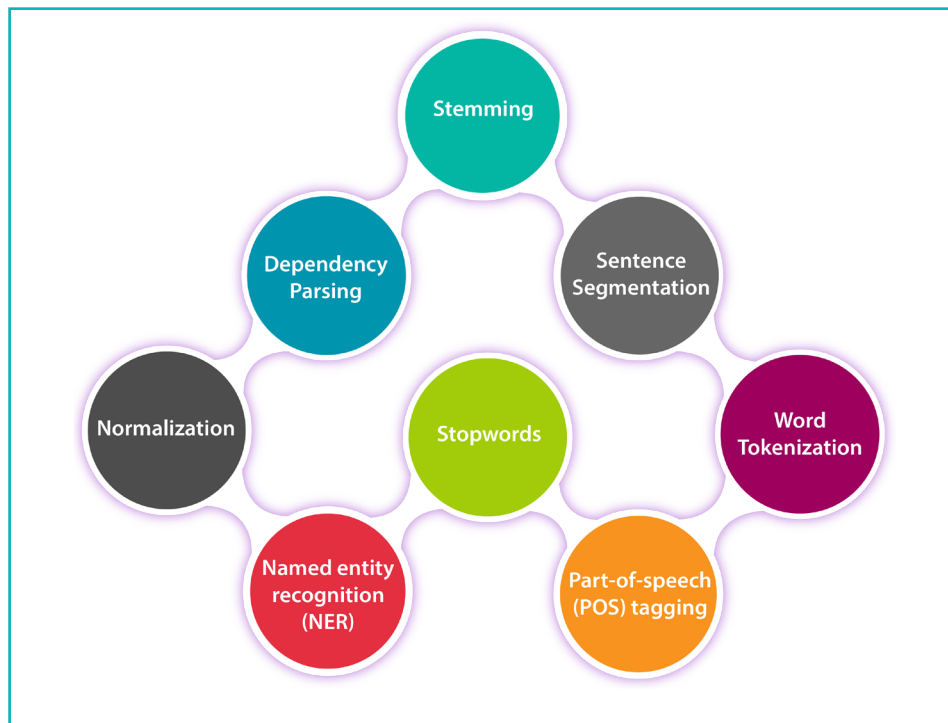
Named Entity Recognition (NER) -This is the process of recognizing and categorizing specific entities in text, such as names of people, places, organizations, and dates. It helps identify and extract important information from the text. For example, if the sentence is 'I want to eat salad,' it will be in the food category.

Stopwords: 'Stopwords' are common words in a language, such as 'a,' 'an,' 'the,' and 'and' that are often removed from texts since they do not carry significant meaning and can be ignored during analysis.

Normalization: It involves transforming text into a standard format that can be easily analyzed and processed. It can include actions such as removing punctuation, converting all letters to lowercase, and replacing numbers with their word equivalents.

Dependency Parsing: This is a technique for analysing the grammatical structure of a sentence by identifying the relationships between words. It helps understand how words depend on or relate to each other in terms of syntax and meaning.

Stemming: 'Stemming' is a technique for converting words into their base or root form. For example, plays, playing, and played, are all derived from the same root word 'play'.



NLP does not perform everything by itself; it has two important components called Natural Language Understanding (NLU) and Natural Language Generation (NLG). Each part has a specific role that helps computers understand, process and interpret human language. Let us explore what each part does and how they work together in NLP.

Components of NLP

NLP and NLG play crucial roles in enabling machines to interact with and comprehend human language in a manner that is both accurate and contextually appropriate. Let us know about them in detail:

NLU (Natural Language Understanding): NLU helps machines to analyze and understand human language. It performs tasks such as figuring out the meaning and emotions behind words, recognizing important information like names and dates, and understanding what someone wants when they ask a question or give a command.

For example, NLU can help determine if a customer review is positive or negative, identify the names of people or places mentioned in a text, or understand that when you ask a virtual assistant to play music, it knows what action to take.

NLU is responsible for the following tasks:

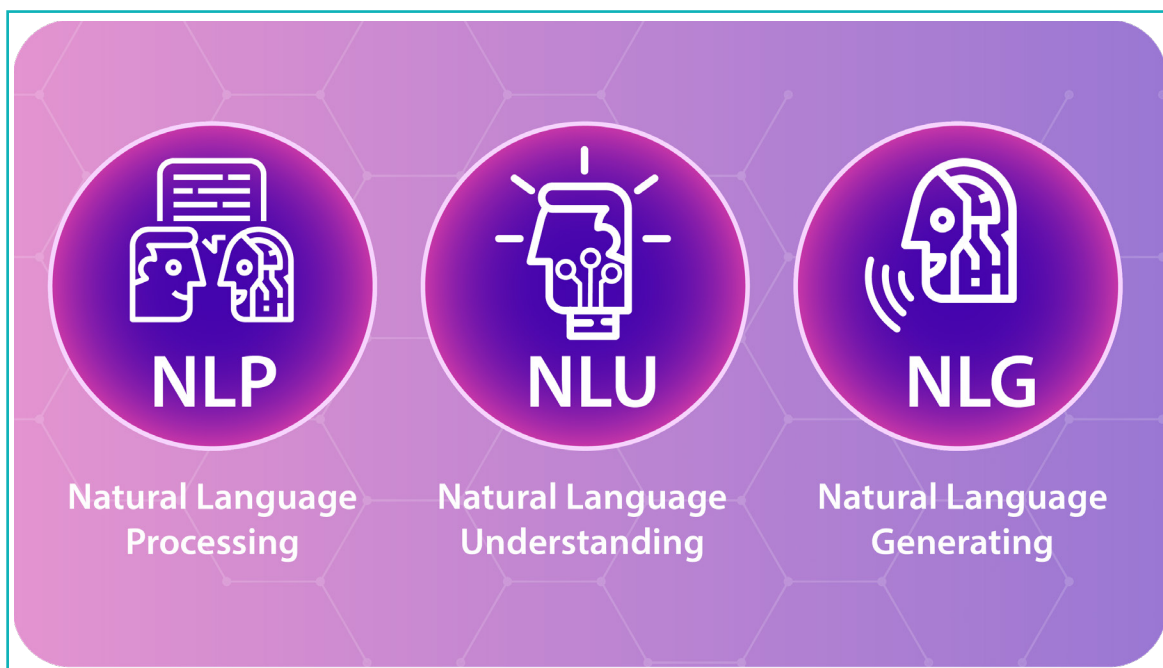
- a. **Sentiment Analysis:** NLU can determine whether a text or speech expresses positive, negative, or neutral feelings. This is helpful for understanding customer feedback, social media posts, and reviews.

- b. Named Entity Recognition (NER):** NER helps NLU identify and categorize specific things like people's names, organization names, locations, dates, and other important information in text or speech. It helps understand the context of the text.
- c. Intent Recognition:** NLU can understand the purpose or what someone wants when they ask a question or give a command. For example, in a virtual assistant, NLU can figure out if someone wants to set an alarm, play music, or search for information.
- d. Information Extraction:** NLU can pick out and gather important information. It uses different techniques to identify and organize specific details mentioned in the text or speech. This information can be used for analysis or other purposes.

NLG (Natural Language Generation) - The goal of NLG is to generate human-like language as output.

NLG is responsible for the following tasks:

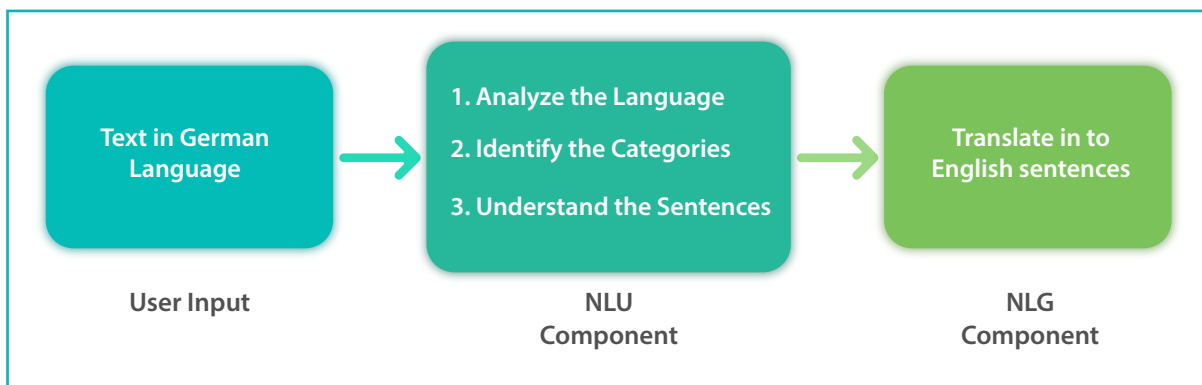
- a. Text Planning:** NLG begins by planning how the text should look. It determines the structure and what information to include. This means deciding how to arrange the paragraphs or sections and creating an outline for the text.
- b. Summarization:** NLG can create short summaries of big texts or documents. It selects the most important information and presents it in a well-organized way.
- c. Language Translation:** NLG is also helpful for translating languages. It can convert text from one language into another, which is really useful for breaking down language barriers.
- d. Dialogue Systems:** NLG is really useful in conversation systems because it helps the machines generate responses while we are talking to them. It tries to make those responses logical and relevant to what we are saying. These dialogue systems that use NLG are used in things like customer support, virtual assistants, and other interactive things we use.



For example,

Suppose a user wishes to translate the input text "Ich möchte heute einen salat essen" from German to English.

The NLU component of the language-translation application will first analyze and understand the meaning of the sentence. It will recognize that the sentence is in German and will identify the word 'Essen' as the action, which means 'to eat'. It will also recognize "Salat" as a food entity, specifically a salad.



After the NLU component has analyzed and understood the input sentence, the NLG component comes into action. It takes the analyzed information and generates a new sentence in English with the same meaning as the original sentence. In this case, the NLG component will produce the sentence "I want to eat salad" in English.

To summarize, the NLU component understands the input sentence and identifies the key elements, such as the language, action, and food entity. Then, the NLG component uses this information to generate a new sentence in the desired language with the same meaning.

Exercise

State if the statements are True or False

5 Text analysis applications, such as Grammarly and Google Docs, are not NLP applications.

 _____

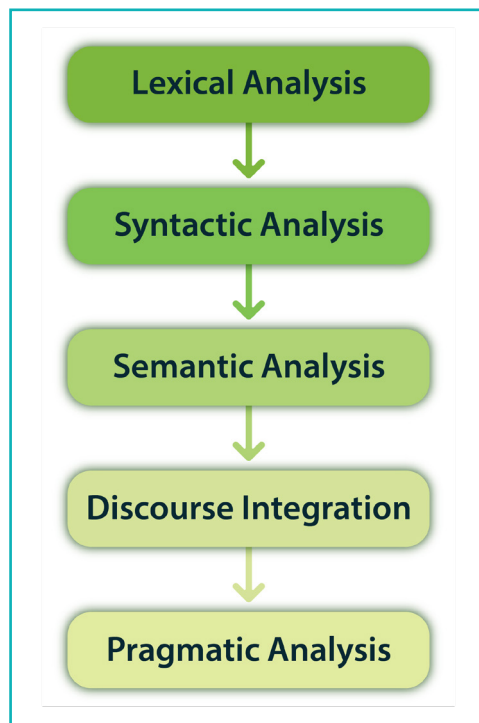
6 Text sentiment is determined during the NLG phase.

 _____

Now, let us look at the various stages of NLP. These stages serve as a road map that guides machines in understanding and generating human language. Each stage has its own purpose and tasks, working together to enable computers to process, interpret, and generate meaningful text and speech.

Stages of NLP

The following are the Natural Language Processing (NLP) processes or stages that are commonly involved in analysing and understanding human language:



Lexical Analysis (Word Analysis) - In this stage, we break down the text into individual words or tokens. Each word is labelled with its respective part of speech, such as noun, verb, adjective, and so on.

For example, in the sentence, 'I am going to the store,' we identify 'I' as a pronoun, 'am' and 'going' as verbs, 'to' as a preposition, and 'the store' as a noun phrase.

Syntactic Analysis (Grammar Analysis) - This stage focusses on understanding the structure of the sentence and the relationships between words. We identify the subject, object, verb, and other elements of the sentence.

For instance, in the sentence 'I am going to the store,' 'I' is the subject, 'am going' is the verb phrase, and 'to the store' is the object.

Semantic Analysis (Meaning Analysis) - Here, we determine the meaning of the sentence by examining the meanings of individual words and their relationships within the sentence.

For example, in the sentence "I am going to the store," the speaker expresses their current location as well as their intention to travel to a specified area.

Discourse Integration Analysis (Context Analysis) - This stage involves understanding the context of the sentence to determine its function and purpose. We consider how the sentence relates to previous or subsequent sentences and the overall conversation.

For instance, the sentence "I am going to the store" might be part of a larger conversation about the speaker's daily routine or their plans for the day.

Pragmatic Analysis (Purpose Analysis) - In this step, we analyze the sentence in terms of its social, political, and cultural context to determine the speaker's intentions and the effect on the listener. We consider the speaker's purpose, whether it is making a request, stating a fact, or conveying an emotion.

For example, the sentence "I am going to the store" could be a request for the listener to accompany the speaker or a simple statement that doesn't require a response.

Exercise

Tick mark ☒ the correct answer

7 Which NLP process includes converting text into a standard format so that it can be easily analyzed?

a. Dependency parsing

☐

b. Normalization

☐

c. Stemming

☐

d. Pragmatic analysis

☐

8 What is the objective of 'Stopwords removal' in NLP?

a. Identifying sentence boundaries

☐

b. Labelling each word with its associated part of speech

☐

c. Recognizing and categorizing named items in the text

☐

d. Removing common words with no relevant meaning

☐

Activity

Problem Statement: Write a Python program to detect the sentiment of text input by the user using an NLP algorithm. Positive, Negative, and Neutral sentiments will be classified.

Before starting the code, let us understand some terms about sentiment.

What is Sentiment?

The overall emotion or attitude expressed in a piece of text, such as a review, social media post, or news article, is known as **Sentiment**.

Types of Sentiment-

Three main types of sentiment: Positive, Negative, and Neutral.

Positive Sentiment: **Positive Sentiment** refers to a favourable or optimistic opinion. For example - This camera takes amazing pictures!

Negative Sentiment: A **Negative Sentiment** is an unfavourable or depressive viewpoint.

For example - This phone battery drains too quickly.

Neutral Sentiment: A lack of emotional opinion is referred to as **Neutral Sentiment**.

For example - This book has 300 pages.



My experience
so far has been
Fantastic!

POSITIVE



The product is
okay I guess.

NEUTRAL



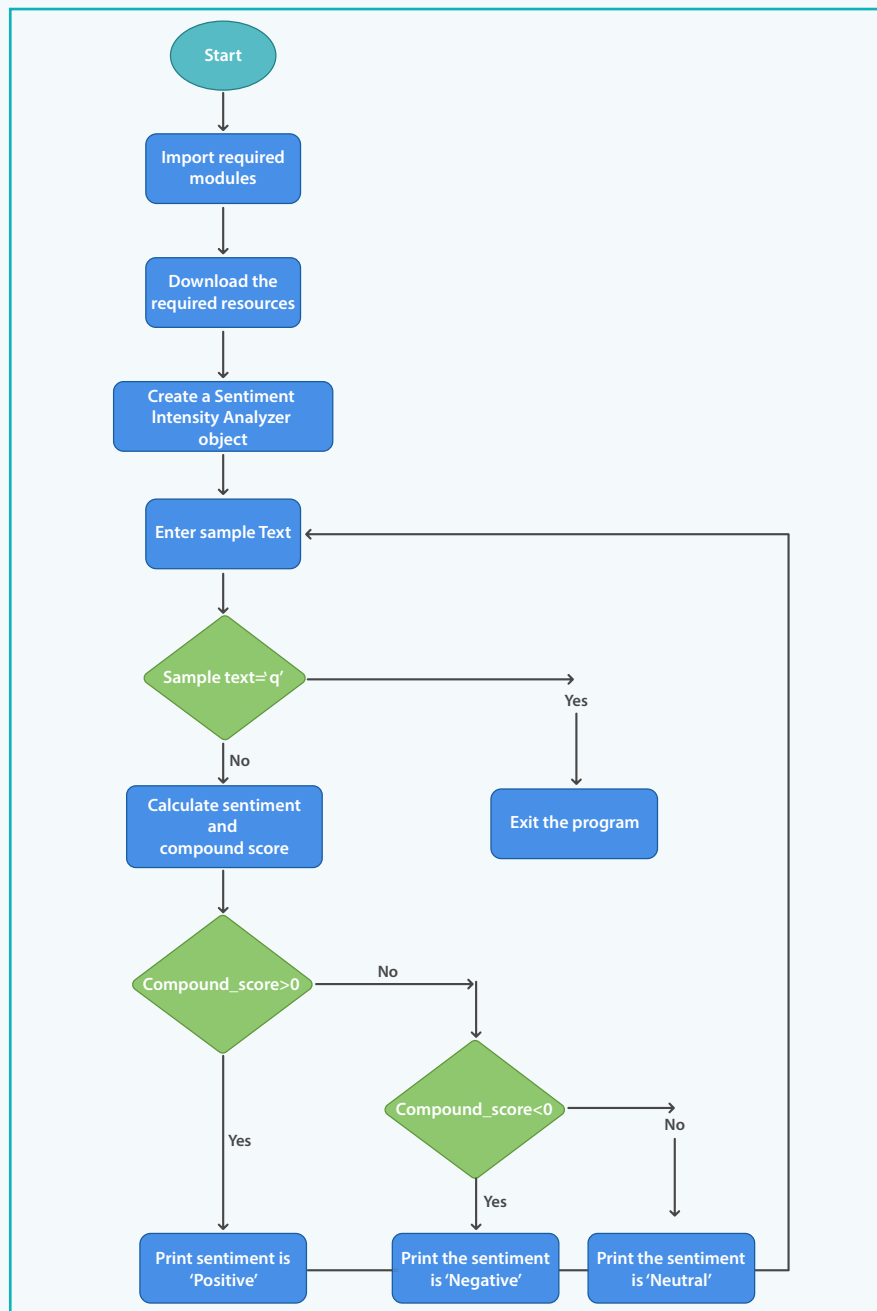
Your Support
team is
useless.

NEGATIVE

Activity

Where is the sentiment analysis used?

Product reviews, social media review, news analysis, customer feedback are some of the applications of sentiment analysis.



Activity

Import required modules and resources

We require the following modules:

nlTK: It is a natural language toolkit used to create Python programmes that interact with natural (human) language data. It includes libraries and programmes for tasks, including tokenization, stemming, tagging, parsing, semantic reasoning, and machine learning.

Sentiment: It is a sentiment analysis package of nlTK module that contains modules and tools for performing sentiment analysis on text.

Vader: Vader (Valence Aware Dictionary and Sentiment Reasoner) is a natural language processing lexicon and rule-based sentiment analysis tool. It analyses text sentiment using a pre-defined collection of sentiment scores for words in the English language.

SentimentIntensityAnalyzer: It is a class provided by the 'nlTK.sentiment.vader' module in NLTK. It uses the Vader sentiment analysis tool to perform sentiment analysis on text.

```
1  #import required module
2  import nltk
3  from nltk.sentiment.vader import SentimentIntensityAnalyzer
4  -
```

- 2 Download the 'vader_lexicon' dictionary. It includes a list of words along with their sentiment scores. The score is in the range of '-4' to '4'.

For example, dictionary is like -

{'determinably': 0.9, 'gloom': -2.6, 'honestly': 2.0, 'honesty': 2.2,}

```
1  #import required module
2  import nltk
3  from nltk.sentiment.vader import SentimentIntensityAnalyzer
4
5  #Download the required resources
6  nltk.download('vader_lexicon')
-
```

Activity

- 3 Create an object of the 'SentimentIntensityAnalyzer' class to access the method that returns the score of words in the input text. Save the object in a variable 'analyzer'.

```
5 #Download the required resources
6 nltk.download('vader_lexicon')
7
8 #Create a SentimentIntensityAnalyzer object
9 analyzer = SentimentIntensityAnalyzer
```

Input the text

- 1 Create an infinite 'while' loop and enter the text using the 'input' function, saving it in the variable 'input_text'.
- 2 If the user types 'q', use the break statement to exit the loop.

```
8 #Create a SentimentIntensityAnalyzer object
9 analyzer = SentimentIntensityAnalyzer
10
11 #Get input text samples from the user
12 while True:
13     input_text = input("\nEnter some text (press 'q' to quit): ")
14     if input_text.lower() == 'q':
15         break
```

Obtaining Score for the Input Text

The 'polarity_scores' method will be used to analyse the sentiment of input text and will return a dictionary containing scores for the text's negativity, neutrality, positivity, and overall compound sentiment. It gives the probability score.

Syntax:

polarity_scores(input text)

Activity

Return:

The 'polarity_scores' method returns a dictionary with the keys and values listed below:

neg: The text's negative sentiment score (a float value between 0 and 1).

neu: The text's neutral sentiment score (a float number between 0 and 1).

pos: The text's positive sentiment score (a float number between 0 and 1).

compound: The text's total compound sentiment score (a float value between -1 and 1, with scores above 0.05 being 'positive', scores below 0.05 being 'negative' and scores between -0.05 and 0.05 being neutral).

Compound Score = Sum of Valence Scores of Each Word / Scaling Factor

- 1 Input the text using 'polarity_scores' method and save the value returned in a variable 'scores'.

```
8 #Create a SentimentIntensityAnalyzer object
9 analyzer = SentimentIntensityAnalyzer
10
11 #Get input text samples from the user
12 while True:
13     input_text = input("\nEnter some text (press 'q' to quit): ")
14     if input_text.lower()=='q':
15         break
16
17     # obtain the score for the input text
18     scores = analyzer.polarity_scores(input_text)
```

- 2 Print the complete sentiment score.

```
11 #Get input text samples from the user
12 while True:
13     input_text = input("\nEnter some text (press 'q' to quit): ")
14     if input_text.lower()=='q':
15         break
16
17     # obtain the score for the input text
18     scores = analyzer.polarity_scores(input_text)
19     #print the sentiment score
20     print("\n Sentiment scores is:-",scores)
21
```

Activity

- 3 Access the value of the compound key from the entire sentiment score, which is in dictionary format, save this score in a variable 'compound_score', and print it.

```
11 #Get input text samples from the user
12 while True:
13     input_text = input("\nEnter some text (press 'q' to quit): ")
14     if input_text.lower()=='q':
15         break
16
17     # obtain the score for the input text
18     scores = analyzer.polarity_scores(input_text)
19     #print the sentiment score
20     print("\n Sentiment scores is:-",scores)
21     compound_score=scores['compound']
22     #print the compound score
23     print("Compound score is:-",compound_score)
24
```

Determine the Sentiment Category

- 1 Based on the value of compound score, determine the category.
- 2 If the score of the compound is greater than zero, the sentiment category is positive; otherwise, the sentiment category is negative; else, it is neutral. Save the sentiment category in a variable 'sentiment'.

```
21     compound_score=scores['compound']
22     #print the compound score
23     print("Compound score is:-",compound_score)
24
25     #Determine the sentiment category
26     if compound_score > 0:
27         sentiment = "Positive"
28     elif compound_score < 0:
29         sentiment = "Negative"
30     else:
31         sentiment = "Neutral"
32
```

Activity

- 3 Print the name of sentiment category.

```
25     #Determine the sentiment category
26     if compound_score > 0:
27         sentiment = "Positive"
28     elif compound_score < 0:
29         sentiment = "Negative"
30     else:
31         sentiment = "Neutral"
32
33     # Print the detected category
34     print("The sentiment of the text is:", sentiment)
35
```

Output

```
Enter some text (press 'q' to quit): My teacher gave me some great feedback on my essay.
Sentiment scores: {'neg': 0.0, 'neu': 0.687, 'pos': 0.313, 'compound': 0.6249}
Compound score: 0.6249
The sentiment of the text is: Positive

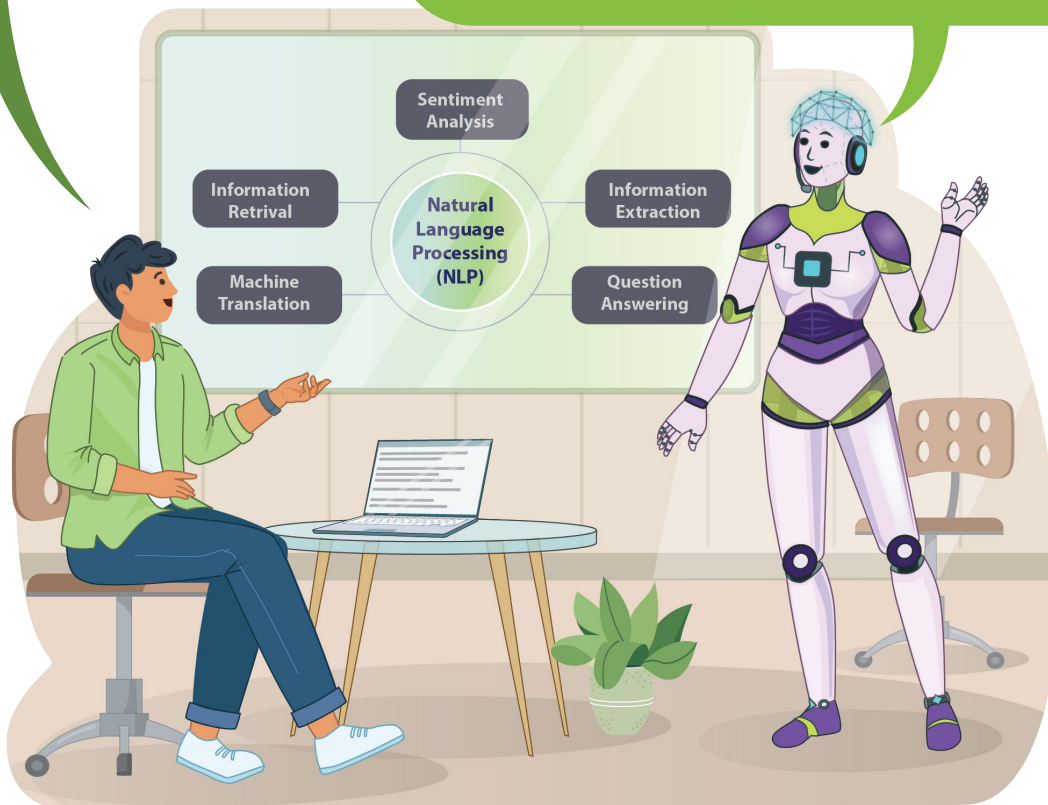
Enter some text (press 'q' to quit): I failed my math test and now my grade is suffering.
Sentiment scores: {'neg': 0.444, 'neu': 0.556, 'pos': 0.0, 'compound': -0.7506}
Compound score: -0.7506
The sentiment of the text is: Negative

Enter some text (press 'q' to quit): I have a lot of homework tonight.
Sentiment scores: {'neg': 0.0, 'neu': 1.0, 'pos': 0.0, 'compound': 0.0}
Compound score: 0.0
The sentiment of the text is: Neutral

Enter some text (press 'q' to quit): q
```

Thank you for sharing your knowledge about NLP and its practical applications. I'm thrilled to apply what I've learned in real-world scenarios.

Fantastic! Sentiment analysis is just the beginning of the vast realm of NLP. Keep exploring different techniques to enhance your skills in this exciting field.



Tech Flash

- **NLP algorithms** : These are computer programs that allow machines to read, understand, and generate human language.
- **NLU and NLG** : NLU is used to analyze and understand human language, while NLG is used to generate human-like language.
- **Lexicon** : In sentiment analysis, a lexicon may include a list of positive, negative, or neutral words and phrases along with their associated scores.